CL13: Sets, Dictionaries, and Intro to Time Complexity

Announcements

- EX03 due *tonight*!
- Submit Quiz 02 regrade requests by 11:59pm tomorrow
- Quiz 03 on Friday
    - Review practice quiz, key, and solutions video
    - Virtual Review Session on Thursday at 6:00pm
    - Please visit Office Hours for help!

Warm-up: Memory Diagram (to submit to Gradescope!)

```python
def group_names(names: list[str]) -> dict[str, int]:
    groups: dict[str, int] = {}
    first_letter: str
    for n in names:
        first_letter = n[0]
        if first_letter in groups:
            groups[first_letter] += 1
        else:
            groups[first_letter] = 1
    return groups

ppl: list[str] = ["Karen", "Emily", "Kris"]
output: dict[str, int] = group_names(names=ppl)
print(output)
output["I"] = 1
print(output)
```

Code-along: Writing a Function that uses a list, set, and dict!

Write a function named `bin_len` that “bins” a list of strings into a dictionary, where the key is an `int` length of a given string and the associated values are a `set` of strings of the key’s length found in the original list.

For example:

`bin_len(["the", "quick", "fox"])` returns `{3: {"the", "fox"}, 5: {"quick"}}`

`bin_len(["the", "the", "fox"])` returns `{3: {"the", "fox"}}`

Before we write any code:

1. What concepts from the course will we need to use?

2. Let’s write this function with pseudocode →

Pseudocode

Python code